

ADVANCES IN HEART FAILURE, MECHANICAL CIRCULATORY SUPPORT AND TRANSPLANT

Combined Post- and Precapillary Pulmonary Hypertension in Patients With Advanced Heart Failure

E. Ashley Hardin¹, MD; Adi Dewanjee², MD; Parker Davis³, MD; Ryan J. Tedford⁴, MD; Nick Hendren⁵, MD; Mark H. Drazner⁶, MD, MSc; Teresa De Marco⁷, MD; Kelly M. Chin⁸, MD

ABSTRACT: Pulmonary hypertension associated with left heart disease, or group 2 pulmonary hypertension, is common in patients with advanced heart failure. In at least a third of these patients, the pulmonary pressures are disproportionately elevated relative to left heart filling pressures and are classified as combined post- and precapillary pulmonary hypertension (CpcPH). CpcPH is thought to result not only from chronically elevated left heart filling pressures but also from neurohormonal activation, a genetic predisposition, and other risk factors. Traditional pulmonary vasodilator therapies are not efficacious and may be harmful in CpcPH. For patients with advanced heart failure, higher-risk CpcPH hemodynamics, particularly significantly elevated or nonreversible pulmonary vascular resistance and transpulmonary gradient, may preclude isolated heart transplantation, while acute vasodilator testing may identify patients who can proceed with acceptable risk. Strategies to minimize posttransplant risk in recipients with CpcPH based on donor selection, for example, oversizing of the donor by relative predicted heart mass or selecting donors with preexisting PH, remain unproven. This review serves to (1) highlight new insights into the pathophysiology and epidemiology of CpcPH; (2) summarize the existing literature surrounding indications for, and interpretation of response to, acute pulmonary vasodilator testing; (3) provide recommendations for management of patients with heart failure and CpcPH, with a particular focus on advanced therapies; and (4) identify knowledge gaps and future research priorities.

Key Words: donor selection ■ heart failure ■ research ■ risk factors ■ vascular resistance

Pulmonary hypertension (PH) associated with left heart disease (LHD) is the most common PH subgroup and affects up to 80% of patients with advanced heart failure (HF).^{1,2} These patients are classified as having World Health Organization group 2 PH and in at least a third of these patients, the degree of elevation in pulmonary pressures is out of proportion to left heart filling pressures, or pulmonary arterial wedge pressure (PAWP), and is subsequently termed combined post- and precapillary pulmonary hypertension (CpcPH). PH-LHD is defined hemodynamically by a mean pulmonary artery pressure of >20 mmHg and PAWP of >15 mmHg and can be further categorized by pulmonary vascular resistance (PVR) into isolated postcapillary PH (PVR ≤ 2 Wood units [WU]) or CpcPH (PVR >2 WU).³

CpcPH impacts eligibility for advanced therapies, particularly heart transplantation. This review serves to (1) highlight a growing understanding of the pathophysiology and prevalence of CpcPH; (2) summarize the existing literature surrounding indications for, and interpretation of response to, acute pulmonary vasodilator testing; (3) provide recommendations for management of patients with CpcPH with a focus on those in need of advanced HF therapies; and (4) identify knowledge gaps and future research priorities.

Pathophysiology of CpcPH

Elevated left atrial pressure in the setting of impaired left ventricular relaxation, with transmission upstream to the

Correspondence to: E. Ashley Hardin, MD, UT Southwestern Medical Center, 5959 Harry Hines Blvd, Ste HP8.110, Dallas, TX 75390-9047. Email elizabeth.hardin@UTSouthwestern.edu

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCHEARTFAILURE.126.014787>.

© 2026 American Heart Association, Inc.

Circulation: Heart Failure is available at www.ahajournals.org/journal/circheartfailure

Nonstandard Abbreviations and Acronyms

AVT	acute vasodilator testing
CpcPH	combined post- and precapillary pulmonary hypertension
HF	heart failure
HFpEF	heart failure with preserved ejection fraction
HFrEF	heart failure with reduced ejection fraction
LHD	left heart disease
LVAD	left ventricular assist device
OHT,	orthotopic heart transplant
PAC	pulmonary artery compliance
PAH	pulmonary arterial hypertension
PAWP	pulmonary artery wedge pressure
PH	pulmonary hypertension
PVR	pulmonary vascular resistance
RV	right ventricular
TPG	transpulmonary gradient
UNOS	United Network for Organ Sharing
WU	Wood units

pulmonary vasculature, underlies the common mechanism of PH in both HF with preserved ejection fraction (HFpEF) and HF with reduced ejection fraction (HFrEF). There is accumulating evidence that the development of pulmonary vasculopathy in patients with HF and CpcPH is not simply due to chronic elevations of left ventricular filling pressures.^{4,5} Why some patients manifest an exaggerated increase in pulmonary pressures out of proportion to PAWP, and others do not, is not clear. Isolated postcapillary PH and CpcPH phenotypes seem to represent a continuum, with the progression from the former to the latter being secondary to a complex interplay of contributing factors.⁶ This transition reflects the cumulative effects and culmination of both the severity and chronicity of elevated PAWP, with resultant vasoconstriction and release of neurohormones and mediators (in particular, endothelin-1) involved in endothelial and smooth muscle cell growth. Other preexisting risk factors for pulmonary arterial vascular disease, including genetic susceptibility, are thought to be superimposed on the pressure load resulting from an elevated PAWP^{6–9} (Figure 1). Histological analyses of pulmonary veins, capillaries, and small arteries from patients with both HFpEF and HFrEF demonstrate significant remodeling of the vasculature, with activation of inflammatory mediators and high levels of urokinase plasminogen receptor expression (for venous remodeling specifically).⁷ Comparative hemodynamic and genetic data show that patients with CpcPH, when compared with those with pulmonary arterial hypertension (PAH), tend to be younger than those with isolated

postcapillary PH and more likely to harbor single nucleotide polymorphisms in genes that are plausibly causative, even after correcting for the duration of HF.⁵

Although commonalities exist between PH due to HFpEF and HFrEF, there are several differences between the 2, suggesting distinct pathobiology.⁷ Patients with PH-HFpEF versus HFrEF had more pulmonary vascular remodeling by DPG and PVR despite a similar PAWP and a lower pulmonary arterial compliance (PAC), suggesting more prominent vascular disease.^{7,10} In addition, patients with HFpEF have a higher burden of metabolic syndrome and obesity, both of which further exacerbate endothelial injury and pulmonary vasculopathy. Further, diastolic stiffness, atrial myopathy, and left atrial functional mitral regurgitation seem to dominate in HFpEF, whereas left ventricular dilation, secondary mitral regurgitation, and left atrial enlargement are more common in PH-HFrEF.⁷ There is a pressing need to better phenotype PH-LHD based on underlying cause (valvular, infiltrative, restrictive, and dilated cardiomyopathies) to improve our understanding of the pathobiology and progression of disease personalized to the specific underlying disease process.²



CpcPH in Patients With Advanced HF

The presence of CpcPH in patients with HF is associated with more severe symptoms, lower exercise capacity, and higher mortality.^{1,11–14} Contemporary series suggest that at least a third of patients with advanced HF and PH manifest a CpcPH phenotype.^{1,13,15} Noncardiac comorbid conditions associated with both HF and the development of precapillary PH may contribute to CpcPH.² It is important to exclude such conditions, including obstructive sleep apnea, obstructive or restrictive lung disease, and chronic thromboembolic PH, as well as drug and toxin exposure or autoimmune disease.³

CpcPH is an important diagnosis in heart transplant candidates, as it may render them ineligible to receive a donor organ. Specifically, a transpulmonary gradient (TPG) >15 mmHg and PVR >5 WU, or a pulmonary artery systolic pressure >60 mmHg along with either elevated TPG or PVR, are relative contraindications to transplantation, as they indicate that the donor right ventricular (RV) may fail posttransplantation.¹⁴ Pretransplant CpcPH is associated with an increased risk of early, severe primary graft dysfunction from RV failure and also with all-cause mortality.^{14,16} Options to improve candidacy for heart transplantation include (1) performing acute vasodilator testing (AVT) for risk stratification; (2) initiation of inotropes or placement of a temporary or durable mechanical support device to lower pulmonary pressures; or (3) combined heart-lung transplantation.

Elevated pulmonary artery pressures without acute reversibility can still decline over days to weeks with maintenance of low left ventricular filling pressures

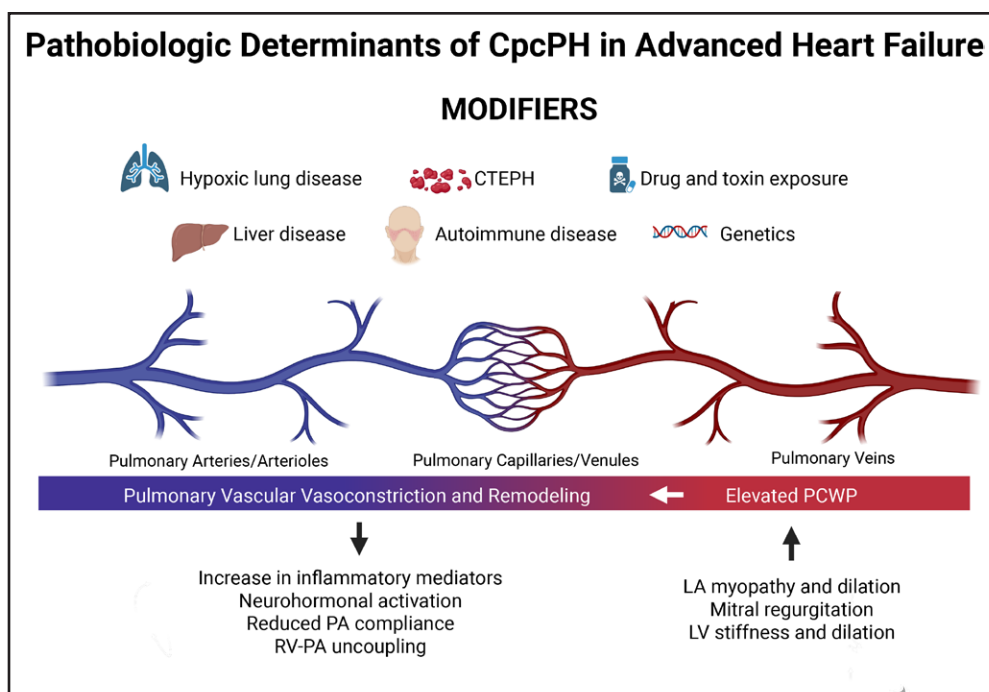


Figure 1. Pathobiologic determinants of combined post- and precapillary pulmonary hypertension (CpcPH) in advanced heart failure.

CpcPH reflects the interaction between elevated left-sided filling pressures and pulmonary vascular vasoconstriction/remodeling, modified by comorbid pulmonary, thromboembolic, hepatic, autoimmune, genetic, and drug/toxin-related factors. CTEPH indicates chronic thromboembolic pulmonary hypertension; LA, left atrial; LV, left ventricular; PA, pulmonary artery; PCWP, pulmonary capillary wedge pressure; and RV-PA, right ventricular-pulmonary arterial. Created in BioRender.com.

following left ventricular assist device (LVAD) implantation as a bridge-to-transplant strategy.^{17,18} While guidelines focus largely on the influence of PH on heart transplant eligibility, CpcPH may also impact outcomes following LVAD implantation due to the consequent physiological and hemodynamic alterations in RV size and function. The increase in RV preload and a reduction in RV contractility/septal shift, with resultant RV-pulmonary artery uncoupling, may increase the risk of post-LVAD right HF.^{1,19}

Pulmonary Vasodilator Testing in CpcPH

The purpose of pulmonary vasodilator testing in patients with CpcPH is to differentiate reversible from fixed PH. This testing is largely used in potential heart transplant recipients to assess posttransplant risk of right HF (Figure 2). However, AVT can also provide risk assessment of RV failure after LVAD placement, as well as an estimate of the overall prognosis in patients with advanced HF.^{20,21}

AVT began with early observations of hemodynamic response to vasodilator infusions in patients with severe mitral valve disease and PH undergoing mitral valve surgery. A subset of these patients demonstrated predominantly vasoreactive, or reversible, PH.^{22–25} In parallel, pulmonary vasoreactivity testing matured in PAH during the 1990s, establishing standardized test agents and

response criteria.^{26,27} Vasodilator testing for a subset of group 1 PAH identifies the small group of patients who would (1) benefit from calcium channel blocker therapies and (2) have a more favorable prognosis compared with other types of group 1 PAH.

The above discoveries led to the uptake of AVT in patients with PH-LHD, though for distinct purposes as compared with when performed in patients with PAH. Transplant programs, confronting excess mortality among patients with elevated pretransplant PVR, adopted AVT during hemodynamic evaluation to identify reversible pulmonary vascular disease and improve candidate selection.^{27,28} These clinical and methodological observations, coupled with accumulating perioperative experience, culminated in guideline recommendations endorsing vasodilator challenge during heart transplant evaluation (Class 1, level of evidence C).¹⁴ Importantly, much of the data on AVT for CpcPH involved patients with HFrEF rather than HFpEF.

Careful attention to detail when obtaining hemodynamic measurements during right heart catheterization as part of AVT is critical to ensure accurate interpretation and conclusions. Guidelines recommend that hemodynamic measurements should be obtained after leveling the transducer at the mid-thorax (approximating the level of the left atrium) and at end-expiration. However, in patients with large intrathoracic pressure swings due to conditions such as obesity and significant

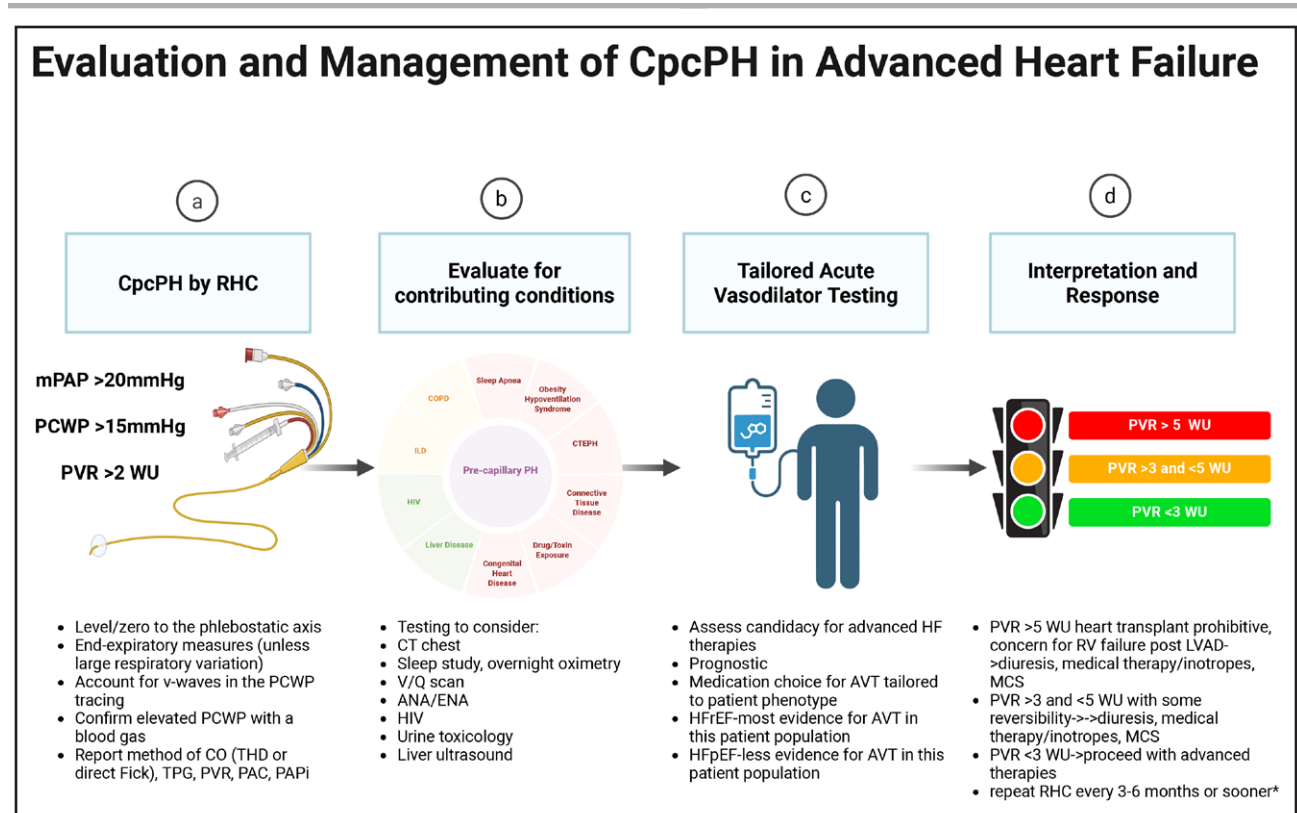


Figure 2. Evaluation and management of combined post- and precapillary pulmonary hypertension (CpcPH) in advanced heart failure (HF).

Proposed framework for the evaluation and management of CpcPH in patients with advanced heart failure. CpcPH is first confirmed by right heart catheterization using careful hemodynamic technique. Patients should then be evaluated for contributing or alternative causes of precapillary pulmonary hypertension. Tailored acute vasodilator testing may then be considered to assess reversibility, candidacy for advanced heart failure therapies, prognosis, and therapeutic phenotype, with the strongest evidence base in HF with reduced ejection fraction (HFrEF) and less established evidence in HF with preserved ejection fraction (HFpEF). Interpretation of the response is based primarily on residual pulmonary vascular resistance, with pulmonary vascular resistance (PVR) >5 Wood units (WU) generally considered prohibitive for isolated heart transplantation and concerning for postleft ventricular assist device (LVAD) right ventricular (RV) failure, PVR 3–5 WU suggesting potentially reversible disease requiring optimization, and PVR <3 WU supporting progression toward advanced heart failure therapies. ANA indicates antinuclear antibody; AVT, acute vasodilator testing; CO, cardiac output; COPD, chronic obstructive pulmonary disease; CTEPH, chronic thromboembolic pulmonary hypertension; ENA, extractable nuclear antigen; ILD, interstitial lung disease; mPAP, mean pulmonary artery pressure; MCS, mechanical circulatory support; PAC, pulmonary artery compliance; PAPI, pulmonary artery pulsatility index; PCWP, pulmonary capillary wedge pressure; PH, pulmonary hypertension; RHC, right heart catheterization; THD, thermodilution; TPG, transpulmonary gradient; and V/Q, ventilation/perfusion. Created in BioRender.com.

lung disease, values averaged over the respiratory cycle may be more appropriate.^{29–31} Importantly, as long as there is consistency in the reporting and interpretation of hemodynamic values with respect to the respiratory cycle (all end-expiratory or all averaged), derived gradients, including TPG and PVR, should be the same. Confirmation of an accurate PAWP by oxygen saturation is essential, as the PAWP influences the derived indices used to assess the degree of precapillary pulmonary vascular involvement. The PAWP should reflect complete pulmonary arterial occlusion. Approximately half of elevated PAWP tracings at first pass reflect incomplete occlusion (as evidenced by PAWP saturation) and require repeat measurement or alternative techniques.³² Large V waves, as seen in severe mitral regurgitation, and a noncompliant left atrium should be accounted for in estimates of the PAWP, as this pressure is transmitted

upstream to the pulmonary arteries during systole.³³ Cardiac output should be obtained by thermodilution or direct Fick methods, the latter being particularly important in heart transplant candidates when pulmonary artery pressures are elevated and the most accurate estimate of PVR is needed, and derived indices such as TPG, PVR, PAC, and pulmonary artery pulsatility index should be reported.

When CpcPH is identified, we strongly advocate that AVT should be performed. The most commonly used agents for AVT in PAH are inhaled nitric oxide and inhaled iloprost, as the inhaled route avoids the risk of hypotension that can occur with systemic vasodilators.³⁴ However, a wider variety of agents are used in clinical practice for AVT in CpcPH, including intravenous milrinone, intravenous sodium nitroprusside, intravenous and sublingual nitroglycerin, inhaled nitric oxide, inhaled

prostacyclins and, in some circumstances, combination testing (Table 1).⁴¹

It is important to note that in patients with elevated PAWP, there is concern that administration of selective pulmonary vasodilatory therapy, such as inhaled nitric oxide, can lead to further increases in PAWP and precipitation of pulmonary edema.^{42,43} These studies highlight the risk of pulmonary vasodilator use in HF, which can unmask limited left-sided reserve and carry adverse prognostic significance. The choice of vasodilator should be tailored to a patient's hemodynamic phenotype (Figure 3). Interpreting the outcome of AVT requires an integrated assessment of blood pressure, PVR, and TPG in concert with changes in PAC, PAWP, and cardiac output; PVR alone is likely insufficient.

Observational studies suggest that patients with CpcPH who do versus do not demonstrate reversible PH with AVT (ie, fixed versus reversible PH), the former defined generally as improvement in PVR to <2.5 to 3 WU and TPG to <12 to 15 mmHg without associated

hypotension (systolic blood pressure <85 mmHg), are suitable candidates for isolated heart transplantation rather than heart-lung transplantation.^{44–47} Identification of recipients with CpcPH who can tolerate isolated heart transplant is important given that there is a substantial benefit, as assessed by long-term survival, when comparing isolated heart transplant versus combined heart/lung transplant.⁴⁸ As such, AVT is recommended before heart transplant in potential recipients with CpcPH. In candidates otherwise eligible for HT with CpcPH, vasodilator testing is particularly informative when PVR is ≥ 3 to 5 WU or the TPG is ≥ 15 mmHg, given the associated increased risks of primary graft dysfunction and mortality with either of these risk markers.¹⁴

Outcomes among patients with a significant vasodilator response before proceeding to isolated heart transplantation have been heterogeneous (Table 2). Some studies have shown relatively similar outcomes across hemodynamic profiles. For example, Drakos and colleagues evaluated 222 patients with reversible PH,

Table 1. Acute Vasodilator Agents for Hemodynamic Testing in Advanced Heart Failure



Vasodilator agent	Mechanism of action	Dosing protocol	Half life	HFrEF	HFpEF	Key physiological effect	Classic patient scenario	Caution with use	Adverse effects
Nitroprusside (IV) ^{30,35}	Balanced arterial/venous vasodilator	Start 0.25–0.5 µg/kg per min; titrate by 0.25–0.5 µg/kg per min every 5 min; max 10 µg/kg per min	2 min	Y	N	↓ afterload + ↓ PAWP → ↑ forward flow	Advanced HFrEF, low or normal CO, elevated PCWP, elevated SVR	SBP <85 mmHg (avoid) Low SVR states Significant renal/hepatic dysfunction	Hypotension Coronary steal Cyanide/thiocyanide toxicity
Nitroglycerin (IV, SL) ³⁶	Venodilator	Start 5 µg/min; titrate by 5 µg/min every 3–5 min; if no response at 20 µg/min, increase by 10–20 µg/min. No fixed max; titrate to hemodynamic response	2–7 min	Y	Y	↓ preload → ↓ PAWP	HFpEF, normal or low CO, elevated PCWP, normotensive	Concomitant nitrates Hypovolemia	Headache Hypotension
Milrinone (IV) ^{35,37}	PDE3 inhibitor (inodilator)	Bolus: 25–75 µg/kg over 1–10 min. Maintenance 0.375–0.75 µg/kg per min	2 h	Y	Select cases without hypotension	↓ PVR + ↓ SVR + ↑ inotropy → ↑ CO	HFrEF, low CO, normal or elevated PAWP, borderline SVR	Significant renal dysfunction Tachyarrhythmias	Hypotension Arrhythmias
Inhaled prostacyclin (iloprost) ^{38,39}	Pulmonary vasodilator	2.5–5 µg via nebulization over 5–10 min. No incremental titration required for acute challenge	Iloprost 20–30 min	Select cases (PAWP <12 mmHg)	Select cases (PAWP <12 mmHg)	↓ PVR (pulmonary-selective), usually with minimal systemic BP effect due to route of administration	HFrEF/HFpEF with PAWP <12 mmHg	Elevated PAWP	Flushing Headache Jaw pain V/Q mismatch Pulmonary edema Hypotension (uncommon)
Inhaled nitric oxide ⁴⁰	Pulmonary vasodilator	10–15 s	10–40 ppm (most centers use 20 ppm); administer for 5–10 min	Select cases (PAWP <12 mmHg)	Select cases (PAWP <12 mmHg)	↓ PVR + ↑ forward flow	HFrEF/HFpEF with PAWP <12 mmHg	Elevated PAWP	Pulmonary edema

Summary of commonly used vasodilator agents for acute hemodynamic assessment, including mechanism of action, dosing, physiological effects, patient phenotype considerations, cautions, and adverse effects. CO indicates cardiac output; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; IV, intravenous; PAWP, pulmonary artery wedge pressure; PDE3, phosphodiesterase-3; PVR, pulmonary vascular resistance; SBP, systolic blood pressure; SL, sublingual; SVR, systemic vascular resistance; and V/Q, ventilation/perfusion.

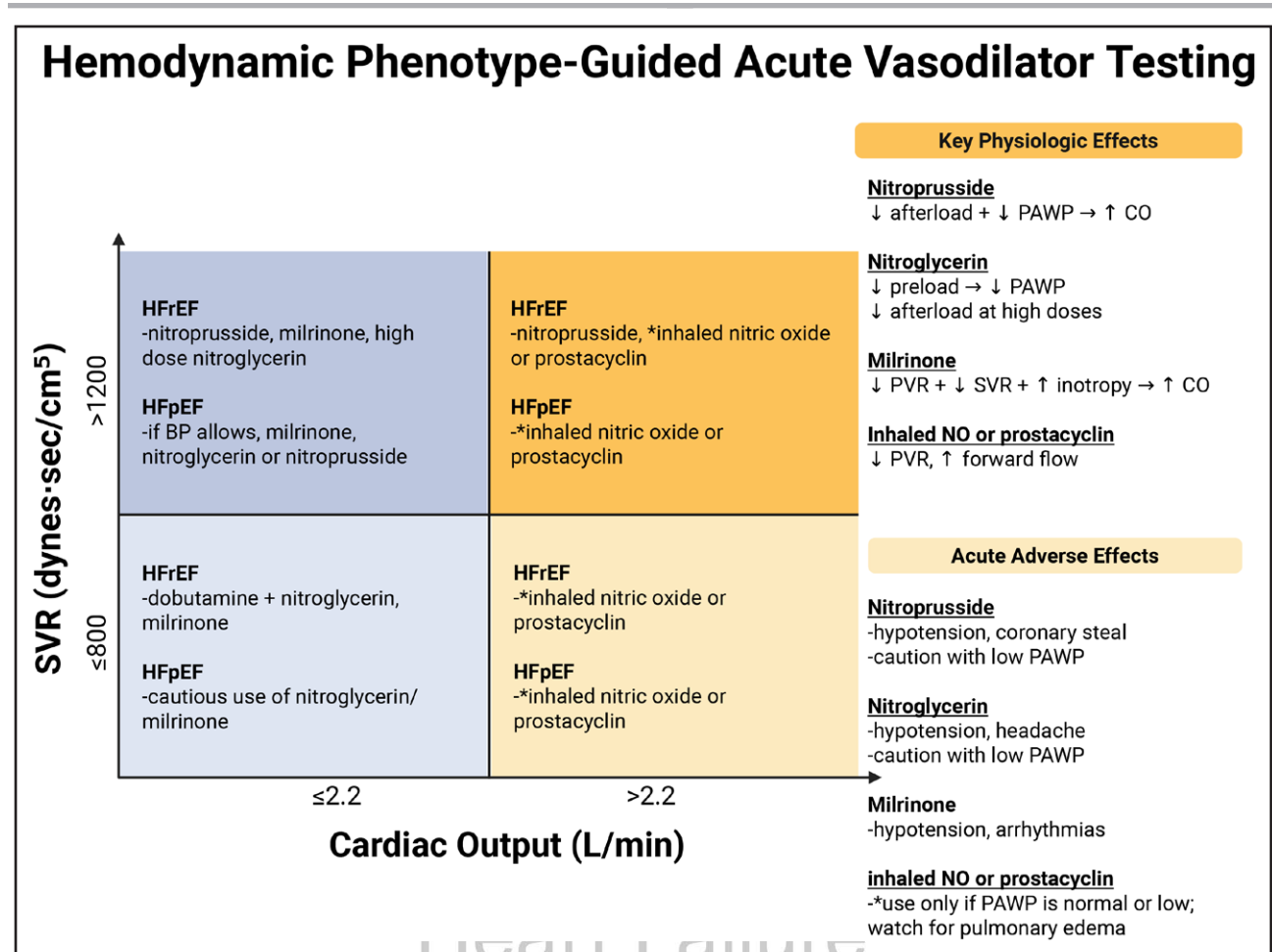


Figure 3. Hemodynamic phenotype-guided acute vasodilator testing (this figure was generated based on author recommendations).

Selection of acute vasodilator testing (AVT) agents may be guided by cardiac output (CO), systemic vascular resistance (SVR), and pulmonary arterial wedge pressure (PAWP), with different considerations for heart failure with reduced ejection fraction (HFrEF) and heart failure with preserved ejection fraction (HFpEF) phenotypes and attention to agent-specific physiological and adverse effects. BP indicates blood pressure; NO, nitric oxide; and PVR, pulmonary vascular resistance. Created in BioRender.com.

stratified by baseline PVR <3 WU versus 3 to 4.5 WU versus >4.5 WU.⁴⁵ Survival at 30 days and 1 year was similar across strata. In another study involving 410 OHT recipients, where pretransplant vasodilator testing defined reversibility, 30-day mortality, intensive care unit length of stay, and long-term survival up to 10 years were similar among patients with no PH, mild-to-moderate PH, and severe PH, provided PVR was reversible or partially reversible to below 6 WU.⁴⁴ Finally, a more recent single-center study, with findings supplemented by a 2014 to 2018 United Network for Organ Sharing (UNOS) analysis, reported no differences in short- or longer-term survival among recipients who were AVT responders when stratified by baseline PVR >3 WU or TPG >15 mm Hg.⁵⁶

In contrast, in a large retrospective cohort spanning 1989 to 2015, pretransplant PVR ≥3.75 WU was associated with higher posttransplant atrial fibrillation and mortality; even responders who reduced PVR to <3.75 WU had lower long-term survival than candidates who began

with lower baseline PVR.⁵⁷ Era effects and confounding, including the use of azathioprine and cyclosporine through the early 2000s and older donor profiles in the high-PVR cohort, may have influenced this study's conclusions. It is unclear how current practices in the modern era of heart transplantation may impact morbidity and mortality of baseline or reversible PH.

PROGNOSTIC IMPLICATIONS OF ACUTE PULMONARY VASODILATOR TESTING AND PREDICTORS OF RESPONSE

Prognostic Implications of Hemodynamics and AVT

In addition to the prognostic value of AVT in patients in need of transplant, AVT is also associated with nontransplant-related outcomes. Patients who have more impaired cardiac function, higher pulmonary

Table 2. Preexisting Pulmonary Hypertension in Patients With Advanced Heart Failure and Outcomes Postheart Transplantation

Study	No. of patients	PVR, TPG	Primary outcome	Results (PH vs no PH)	Results (Rev PH vs no PH)
Costard-Jackle ⁴⁹	293	PVR >2.5 vs PVR <2.5 vs reversible PVR to <2.5 *Nitroprusside	90-d mortality (from RHF/PH)	17.9% (6.7%) vs 6.9% (0%)	40.6% PVR >2.5 no reversibility 27.5% PVR >2.5, reversible but SBP <85 3.8% PVR >2.5 but reversible and SBP >85 6.9% PVR <2.5
Murali ⁵⁰	425	PVR >5 and TPG >15	30-d mortality from primary graft failure	15% vs 4.2%	
Delgado ⁵¹	112	PVR >2.5 and TPG >12	30-d mortality	24.4% vs 5.6%	
Klotz ⁵²	151	PVR >2.5 and TPG >12 vs reversible PVR <2.5 *PGE1	1 y mortality		22% vs 14% (ns, small numbers who went to OHT)
Klotz ⁵³	217	PVR >2.5 and TPG >12, reversible with PGI2 or PGE1	30-d RHF 30-d survival 10-y survival	64% vs 27% 78% vs 85% (ns) 61% vs 63% (ns)	22% vs 14% (ns, small numbers who went to OHT)
Goland ⁴⁴	410	PVR >3 and TPG >10 mild/mod PH PVR 3–6, TPG 10–20 severe PH PVR > 6 and TPG >20 (*reversible to proceed with OHT)	30-d mortality		No PH (266) 2.7% vs mild/mod (112) PH 2.8% vs severe PH (32) 3.2% (ns)
Butler ⁵⁴	182	PVR >2.5 reversible to <2.5 vs normal PVR *Nitroprusside, iNO	Mortality (avg follow up 52 mo)		PVR 2.5–4 OR 1.72 PVR >4 OR 2.35
Drakos ⁴⁵	222	PVR <3 PVR >3–4.5 PVR >4.5 *Reversible to proceed with OHT	<30 d mortality 1-y mortality		2 vs 0 vs 13% (ns, PVR <3 vs 3–4.5 vs >4.5) 8 vs 0 vs 13%
Vakil ⁵⁵	UNOS 26 649	Pretransplant PH Severe PH PVR >5, TPG >20	Overall mortality (severe vs no PH)	(30 d) HR 1.28 (6 mo) HR 1.14 (1 y) HR 1.09 (5 y) HR 0.98 (10 y) HR 0.94	
Crawford ¹¹	UNOS 32 465	PVR >3 vs PVR <3 *True even when mPAP <25	30-d mortality	HR 1.16	

HR indicates hazard ratio; iNO, inhaled nitric oxide; ns, nonsignificant; OHT, orthotopic heart transplantation; OR, odds ratio; PGE1, prostaglandin E-1; PGI2, prostaglandin I-2; PH, pulmonary hypertension; PVR, pulmonary vascular resistance; RHF, right heart failure; SBP, systolic blood pressure; TPG, transpulmonary gradient; and UNOS, United Network for Organ Sharing.

arterial pressures, smaller percent reductions in PVR, or an increase in PAWP following AVT have worse outcomes. Specifically, in a multicenter, retrospective study involving patients with an average PVR of 5.3 WU, peak stroke volume index postsodium nitroprusside administration predicted RV reserve and risk for post-LVAD RV failure.^{20,21} In another single-center study involving 54 patients undergoing VAD implantation (average PVR 4 WU), postsodium nitroprusside infusion changes in RV fractional area change, pulmonary artery systolic pressure, and pulmonary artery pulsatility index were associated with post-LVAD RV failure.²¹ In addition, AVT is prognostic for HF-related morbidity and mortality in general. In a cohort of 98 patients with advanced systolic HF and CpcPH (defined by either TPG ≥ 12 mm Hg, PVR ≥ 3 WU, or DPG ≥ 7 mm Hg), a $\geq 20\%$ reduction in PVR during sodium nitroprusside infusion was associated with lower risk of the composite end point of death, transplantation, or the need for durable mechanical support.⁵⁸ A smaller retrospective Japanese series of 69 patients with group

2 PH showed that an increase in PAWP during inhaled nitric oxide testing was associated with increased risk of all-cause mortality or hospitalization for HF⁵⁹ (Table S1).

Response to AVT

Several clinical and hemodynamic parameters are associated with nonresponse to AVT. In a large, retrospective analysis of 402 patients with advanced HF and PH (average PVR was 3.61 WU) undergoing vasodilator challenge, low baseline diastolic blood pressure (<70 mm Hg), elevated PVR (>5 WU), and reduced PAC (<1.2 mL/mm Hg) were associated with absence of vasoreactivity. The presence of any 2 of these parameters conferred $\approx 90\%$ probability of nonresponse.⁶⁰ In the PROPHET trial, a multicenter, prospective study involving 208 patients undergoing evaluation for heart transplantation who had PH and underwent IV prostaglandin E1 infusion, higher baseline TPG, absence of COPD, male sex and higher left ventricular ejection fraction were associated with improved

response to AVT.⁶¹ When an integrated response to AVT yields acceptable RV afterload and preserved left-sided filling pressures, pursuing isolated OHT is felt to be acceptable with tailored perioperative planning and management. Multiple strategies are considered to minimize the risk peri-heart transplant including the use of short- and longer-term medical and device-based therapies, as well as nuances in donor selection.

MANAGEMENT STRATEGIES FOR CPCPH

Medical and Device Management

In current practice, aggressive decongestion, guideline-directed medical therapy, afterload reduction, inotropes, temporary or durable mechanical circulatory support, and selective short-term pulmonary vasodilators in monitored settings may be administered to patients with marked CpcPH. In heart transplant candidates, treatment strategies should be individualized to the patient and include considerations regarding whether euvolemia can be maintained, the degree of hemodynamic derangements and anticipated waitlist time (Figure 4). Pulmonary hemodynamics should be repeated following interventions before listing to demonstrate improvement, and then serially every 3 to 6 months.^{35,62} For those patients who have a favorable response to AVT, there is no consensus on how this vasoreactivity might influence the choice of pharmacological therapies or timing of repeat hemodynamic assessment. There is a growing body of evidence supporting the use of temporary mechanical support devices, particularly percutaneous transvalvular microaxial flow devices, to lower pulmonary pressures and PVR over days to weeks in those patients with more urgent indications for advanced therapies (Class IIa indication in the most recent guidelines^{14,63–69}). Durable LVAD therapy has long been considered a bridge-to-transplant strategy for patients who have PH prohibitive to transplant. Studies spanning the past 2 decades have consistently demonstrated the efficacy of nonpulsatile, continuous-flow devices (both axillary and centrifugal) in substantially reducing pulmonary pressures in the majority of patients over a period of months.^{70–77} In a study of 85 patients who underwent LVAD therapy, 25 of whom had an average PVR ≥ 4 WU, PVR improved significantly in those with baseline elevation, and post-LVAD PVR measurements were similar in those with and without elevation at baseline.⁷³ In addition, in a study involving 145 patients who underwent LVAD implantation for fixed PH, 27 had PH as the only contraindication to transplant (PVR 3.49 ± 1.47 WU pre-LVAD) and subsequently underwent repeat catheterization. A PVR < 2.5 WU was achieved in 89% (PVR 1.53 ± 0.66 WU post-LVAD), and 19 were able to undergo heart transplantation.⁷⁶

The medical management of CpcPH is challenging given the limited efficacy and potential harm observed

with chronic PAH-targeted therapies in PH-LHD (Table 3). The most common adverse effects of these agents in patients with HF have included volume overload and an increase in HF hospitalizations.^{78,80–82} The lack of clinical efficacy of these agents likely reflects a mismatch between the pathobiology and the mechanism of action of traditional PAH therapies. Histopathologic analyses of autopsy specimens from patients with HF and PH-LHD, irrespective of ejection fraction, display global pulmonary vascular remodeling with prominent venous and small-vessel intimal thickening, with a histological pattern similar to, but generally less severe than, that in patients with pulmonary veno-occlusive disease.⁸⁶ PH severity appears to correlate more strongly with venous intimal thickening rather than medial or intimal thickening of pulmonary arteries. Lowering arteriolar tone in the face of left-sided cardiomyopathy, left atrial myopathy and pulmonary venous remodeling risks worsening pulmonary congestion without materially improving effective RV afterload, particularly in HFpEF.

Although chronic PAH-directed therapies in group 2 PH have not been beneficial, whether short-term pulmonary vasodilators administered to patients with CpcPH awaiting heart transplant, especially those with a favorable AVT response, improve outcomes remains unknown; overall, the evidence base is sparse.¹ Small, single-center, nonrandomized studies demonstrated benefit of PDE5i therapy in patients with CpcPH in need of heart transplantation.^{87–89} However, no prospective trial has evaluated this strategy as a bridge to transplant despite single-center experiences. Guidelines do not provide specific recommendations for or against PDE5i in CpcPH, stating that an individualized approach is warranted and that harmful effects cannot be excluded.³⁰ Sotatercept, a first-in-class activin signaling inhibitor approved for PAH in 2024, is an attractive potential therapy in CpcPH because it targets pulmonary vascular remodeling while also possibly having decongestive effects. In the recently published phase 2 CADENCE trial, sotatercept reduced PVR compared with placebo in patients with HFpEF and CpcPH.⁹⁰ However, the trial excluded many patient phenotypes relevant to transplant programs, such as those with infiltrative/obstructive cardiomyopathies, significant valvular heart disease, and HFpEF.

Guideline-directed medical therapy has an emerging role in patients with PH and RV failure. SGLT2 inhibitors may have beneficial hemodynamic effects in the pulmonary vasculature. Several small trials demonstrated meaningful improvements in pulmonary pressures with SGLT2 inhibitor therapy, independent of their diuretic effect. The EMBRACE-HF trial (n=65) demonstrated that empagliflozin significantly reduced pulmonary artery diastolic pressures in patients with mildly reduced and preserved ejection fraction, after just 1 week of treatment, as assessed by implanted pulmonary artery pressure sensors (CardioMEMS).⁹¹ The CAMEO-DAPA trial

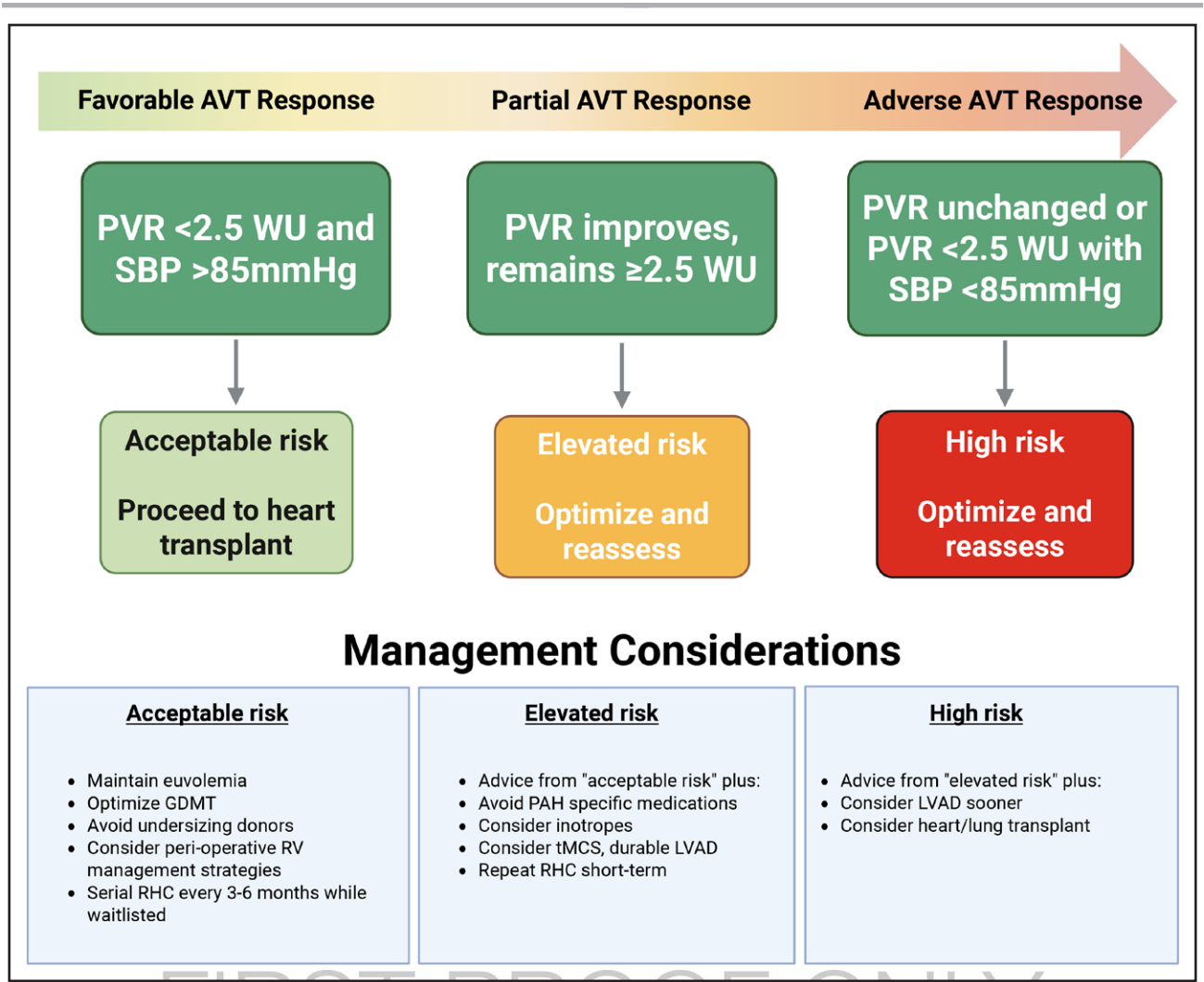


Figure 4. Interpretation and management after acute vasodilator testing (AVT; author recommendations based on expert consensus). AVT response stratifies patients into favorable, partial, or poor/adverse response categories based on pulmonary vascular resistance (PVR) and systolic blood pressure (SBP), informing transplant candidacy, optimization strategy, timing of left ventricular assist device (LVAD)/temporary mechanical circulatory support (tMCS), and need for repeat right heart catheterization (RHC). GDMT indicates guideline-directed medical therapy; PAH, pulmonary arterial hypertension; RV, right ventricular; and WU, Wood units. Created in BioRender.com.

(n=38) demonstrated significant decreases in pulmonary artery wedge, right atrial, and mean pulmonary pressures compared with placebo.⁹² Further, preclinical data corroborate these effects. Canagliflozin led to dose-dependent pulmonary artery relaxation in a murine model.⁹³ Likewise, SGLT2 inhibitor treatment in a rat model of PAH improved survival, reduced mean pulmonary artery pressure and RV systolic pressure, and reduced RV hypertrophy and fibrosis.⁹⁴ In addition to direct effects on the vasculature, SGLT2 inhibitors may promote a favorable hemodynamic milieu via osmotic diuresis, improved insulin sensitivity and reduced inflammation.⁹⁵

Angiotensin receptor-neprilysin inhibitor therapy has also been shown to improve right heart function, particularly among individuals with reduced ejection fraction. In a meta-analysis of 10 observational studies of individuals with HFpEF, sacubitril-valsartan improved several

indicators of right heart function, including tricuspid annular plane systolic excursion, tricuspid annular peak systolic velocity, and pulmonary artery systolic pressure.⁹⁶ Among individuals with PH related to HFpEF, angiotensin receptor-neprilysin inhibitors have also demonstrated promise. In 2 recent studies, treatment with sacubitril-valsartan reduced PAWP and mean pulmonary artery pressure.^{97,98} Notably, rebound of mean pulmonary artery pressure was observed after withdrawal of the angiotensin receptor-neprilysin inhibitor.⁹⁸

Donor Selection for Heart Transplant Candidates With CpcPH

Historically, it has been customary to oversize donor hearts for recipients with significant PH in the hope that a large right ventricle would mitigate the risk of

Table 3. Pulmonary Vasodilator Therapy in Patients With Heart Failure

Medication class	Trial name	First author	Year	N	Intervention	Inclusion criteria	Results
PDE3 inhibitors	PASSION ⁷⁸	Hoeper	2024	125	Tadalafil vs placebo	LVEF $\geq 50\%$, mPAP ≥ 25 , PVR ≥ 3 , PAWP ≥ 15	No improvement in event-free survival, time to HF-death, NYHA FC, 6MWD, NT-proBNP, QOL
	Sil-HF ⁷⁹	Cooper	2022	69	Sildenafil vs placebo	LVEF $< 40\%$, PASP ≥ 40 by TTE	No improvement in symptoms, QOL, PASP, or 6MWD
	SIOVAC ⁸⁰	Bermejo	2018	200	Sildenafil vs placebo	mPAP ≥ 30 , prior valve replacement (mitral and aortic)	Worse outcomes in treatment arm at 6 mo (composite score of death, HF hospitalization, WHO class, and self-assessment)
Endothelin receptor antagonists	SER-ENADE ⁸¹	Shah	2025	142	Macitentan vs placebo	LVEF ≥ 40 , PVR ≥ 3 OR DPG ≥ 5 OR mPAP ≥ 25 OR (TTE PASP ≥ 40 and RV dysfunction)	No effect on NT-proBNP or KCCQ but higher rate of fluid retention and adverse cardiac events with macitentan
	MELODY ⁸²	Vachiéry	2018	63	Macitentan vs placebo	mPAP ≥ 25 , PAWP > 15 and < 25 , PVR ≥ 3 and DPG ≥ 7	No change in PVR, RAP, or PAWP
	ENABLE (1 and 2) ⁸³	Packer	2017	1613	Bosentan vs placebo	LVEF $\leq 35\%$	No significant clinical change. Increased risk of early HF hospitalization. No change in mortality
Guanylate cyclase	HemoDYNAMIC ⁸⁴	Dachs	2022	114	Riociguat vs placebo	LVEF $\geq 50\%$, mPAP > 25 , PAWP > 15	Significant improvement in CO, decrease in PVR, TPG. No change in NT-proBNP, exercise capacity. Favorable safety profile
$\beta 3$ adrenergic agonist	SPHERE-HF ⁸⁵	García-Álvarez	2023	80	Mirabegron vs placebo	mPAP ≥ 25 , PAWP or LVEDP > 15 , PVR ≥ 3 or DPG ≥ 7	No change in PVR, improved RVEF in treatment group

All hemodynamic measurements reported in mmHg. 6MWD indicates 6-minute walk distance; CO, cardiac output; DPG, diastolic pulmonary gradient; HF, heart failure; KCCQ, Kansas City Cardiomyopathy Questionnaire; LVEDP, left ventricular end-diastolic pressure; LVEF, left ventricular ejection fraction; mPAP, mean pulmonary artery pressure; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA FC, New York Heart Association functional classification; PASP, pulmonary artery systolic pressure; PAWP, pulmonary artery wedge pressure; PVR, pulmonary vascular resistance; QOL, quality of life; RAP, right atrial pressure; RV, right ventricular; RVEF, right ventricular ejection fraction; TPG, transpulmonary gradient; TTE, transthoracic echocardiogram; and WHO, World Health Organization.

posttransplant morbidity and mortality. However, few data support the benefit of this strategy. A retrospective review of the UNOS database demonstrated that the main risk comes with undersizing donors in this recipient population rather than a benefit from oversizing the donor heart. Tao and colleagues⁶⁴ evaluated over 37 000 patients in the UNOS database from 2003 to 2022. Notably, over 20% of recipients had a pretransplant PVR of at least 3 WU. Predicted heart mass $< 15\%$ versus that of the recipient (ie, undersizing) was associated with a significantly higher risk of posttransplant death, with a graded risk based on the recipient's PVR. In contrast, oversizing (predicted heart mass $\geq 15\%$ of the recipient predicted heart mass) was not necessarily beneficial. Whether response to vasodilator testing might alter these findings is unknown. Another analysis of a UNOS cohort from 1994 to 2016 included over 29 000 patients, 36% of whom had moderate PH as defined by a PVR of 2.5 to 5 WU or TPG of 10 to 19 mmHg, and showed comparable results: undersizing was associated with increased risk in recipients with elevated PVR, but oversizing did not confer an advantage even when accounting for female donors to male recipients.⁹⁹

Another strategy to minimize risk of heart transplantation in recipients with CpcPH is to select donors who have preexisting PH, with the hypothesis being that their right ventricles would be preconditioned to tolerate the expected PH following transplantation. However, limited

data are available to support this practice. In a UNOS cohort analysis from 2010 to 2023, higher donor pulmonary pressures did not impact postheart transplant 30-day or 1-year survival in adult recipients.¹⁰⁰ This too remains an area in need of further study.

CONCLUSIONS AND CALL TO ACTION

CpcPH exists in a significant proportion of patients in need of advanced HF therapies. The pathophysiology is complex and reflects a combination of severity and duration of elevated left-sided filling pressures, along with neurohormonal mediators and a background of genetic predisposition to disease. Patients with advanced HF and higher-risk or nonreversible CpcPH hemodynamics pose unique challenges because of increased perioperative risk with heart transplantation and potential risk of right HF after LVAD implantation. AVT may help identify patients who have an acceptable risk for surgical HF therapies. However, there remain important questions that need further study (Table 4).

Current guidelines for heart transplant candidates with CpcPH are based on largely historical data. It is unknown whether interval changes in the field of heart transplantation, such as allocation reforms, broader geographic sharing, increased reliance on temporary mechanical support, rapid uptake of donation after cardiac death allografts, and advanced preservation strategies, should impact these recommendations. In particular,

Table 4. Key Unanswered Questions and Future Approaches

Topic	Key question	Study approach	Proposed Primary end points
AVT response thresholds	Are traditional AVT cut points (PVR ≥3–5 WU; TPG ≥15) valid with post-2018 allocation, DCD use, and novel preservation techniques?	Prospective multicenter registry with core-laboratory hemodynamics; stratify by DBD vs DCD and preservation method	PGD at 30 d, RVAD/ECMO, 1-y survival, early post-HT PVR trajectory
Short-term vasodilators pre-HT	Do protocolized iNO/prostacyclin/SNP/NTG courses improve waitlist and peri-op outcomes vs optimized usual care?	Pragmatic RCT before listing	Waitlist mortality/delisting, peri-op RV failure, 90 d and 1-y survival
Pre-AVT optimization	Does standardized decongestion/inotrope/tMCS sequencing before AVT change “reversibility” classification and outcomes?	Cluster randomized care-pathway vs usual care; core-laboratory AVT review	Post-HT RV failure, HT vs LVAD strategy outcomes
Donor strategy: DCD, PHM matching	Do DCD grafts or PHM oversizing rules differ for high-PVR recipients? Is “preconditioned” donor RV relevant?	Registry with interaction testing (DCD vs DBD × PHM match × recipient PVR)	PGD, ICU LOS, 1-y survival
Waitlist priority policy	Should CpcPH with PVR of 3–5 WU (esp. HFpEF/RCM) receive higher priority to prevent progression to fixed PH?	Decision-analytic modeling with SRTR/UNOS validation	Waitlist mortality, time to HT, post-HT outcomes
AVT response metric	Is a composite (PVR + PAC + ΔPAWP + BP tolerance) superior to PVR alone for predicting peri-op risk?	Prospective core-laboratory AVT classifier comparison	Net reclassification for PGD; 1-y survival
Strategy for nonresponders	LVAD-first vs intensified optimization then HT-what maximizes survival and PH reversal?	Prospective RCT with serial hemodynamics	Overall survival, RV failure, time to acceptable PVR
Phenotype and genetics	Do thresholds and AVT interpretation differ by phenotype (HFpEF/RCM vs HFrEF) and genetic background (eg, EIF2AK4)?	Prospective registry + genetic substudy	PGD, survival; association of variants with AVT response

AVT indicates acute vasodilator testing; BP, blood pressure; CpcPH, combined post- and precapillary pulmonary hypertension; DBD, donation after brain death; DCD, donation after circulatory death; ECMO, extracorporeal membrane oxygenation; HFpEF, heart failure with preserved ejection fraction; HT, heart transplantation; iNO, inhaled nitric oxide; LOS, length of stay; LVAD, left ventricular assist device; NTG, nitroglycerin; PAC, pulmonary arterial compliance; PAWP, pulmonary artery wedge pressure; PGD, primary graft dysfunction; PHM, predicted heart mass; PVR, pulmonary vascular resistance; RCM, restrictive cardiomyopathy; RCT, randomized controlled trial; RV, right ventricle; RVAD, right ventricular assist device; SNP, sodium nitroprusside; SRTR, Scientific Registry of Transplant Recipients; tMCS, temporary mechanical circulatory support; TPG, transpulmonary gradient; UNOS, United Network for Organ Sharing; and WU, Wood units.

the applicability and prognostic significance of AVT need to be reassessed in the modern era.

other pharmacological therapies with novel mechanisms of action should be tested.

Two priorities are paramount. First, multicenter prospective registries should establish uniform AVT protocols with core laboratory adjudication of hemodynamic tracings, prespecified end points that include PVR, TPG and DPG, PAC and RC-time, and standardized reporting of cardiac output methodology. The registries should include patients with infiltrative cardiomyopathies and HFpEF in addition to HFrEF. Their findings should be reported stratified by HFpEF versus HFrEF, as well as CpcPH versus isolated postcapillary PH. Vasodilator agents used, and their doses, should be reported. Second, era-appropriate thresholds for OHT candidacy require validation; the conventional cut points of ≥3 and ≥4 to 5 WU, which should always be interpreted in the context of overall hemodynamic parameters, need to be tested against outcomes in the post-2018 allocation and donation after cardiac death environment. AVT response and PAC, rather than PVR alone, should be considered as hemodynamic risk predictors. Comparative studies of LVAD-first versus intensified medical optimization for AVT nonresponders should include posttransplant hemodynamics and survival as end points. Therapeutic developments in CpcPH should target venous and venular remodeling, left atrial myopathy, and endothelial-metabolic pathways, especially in HFpEF-enriched phenotypes, with careful attention to perioperative safety in candidates likely to require advanced therapies. Finally, the role of sotatercept and

ARTICLE INFORMATION

Received June 17, 2026; accepted August 6, 2026.

Affiliations

UT Southwestern Medical Center, Dallas, TX (E.A.H., A.D., P.D., N.H., M.H.D., K.M.C.); Medical University of South Carolina, Charleston (R.J.T.). University of California San Francisco (T.D.M.).

Sources of Funding

None.

Disclosures

Dr Tedford's general disclosures include consulting relationships with and receiving honoraria/consulting fees from Abbott (active), Acorai (active), Adona (active), Aria CV, Inc (active), Boston Scientific (inactive), CVRx (inactive), Endotronix (active), Edwards Lifesciences (active), Fauna Bio (inactive), Gradient (active), Imbria (inactive), Medtronic (active), Merck (active), Mitre (active), Morphic Therapeutics (inactive), Pulmovant (inactive), Restore Medical (active), Tempus AI (active), 35Pharma (active), and United Therapeutics (active). Dr Tedford serves on steering committees for Abbott, Edwards, Endotronix, Gradient, Merck, Restore Medical, and Tempus AI, as well as a research advisory board for Abiomed. Dr Hendren has received fees from Tosoh, Inc, for clinical trial adjudication. Dr De Marco has received fees for consulting or advisory work from Actelion, Janssen, Johnson & Johnson, Alexion, Aerovate, Boston Scientific, Corsair, IQVIA-Sanofi, Imricor, Kamada, Keros, Merck, Pulnova, Natera and United Therapeutics. Her institution has received support from CareDx. Dr Chin has received fees for work on steering, advisory, or adjudication committees with Gossamer Bio, Inhibikase, Janssen, Merck, and United Therapeutics, and her institution has received research support for clinical studies overseen by her from Gossamer Bio, Janssen, Merck, Novartis, Pulmovant, and United Therapeutics. The other authors report no conflicts.

Supplemental Material

Table S1
References 20,21,58,59

REFERENCES

- Sparrow CT, LaRue SJ, Schilling JD. Intersection of pulmonary hypertension and right ventricular dysfunction in patients on left ventricular assist device support: is there a role for pulmonary vasodilators? *Circ Heart Fail*. 2018;11:e004255. doi: 10.1161/CIRCHEARTFAILURE.117.004255
- Maron BA, Bortman G, De Marco T, Huston JH, Lang IM, Rosenkranz SH, Vachiéry J-L, Tedford RJ. Pulmonary hypertension associated with left heart disease. *Eur Respir J*. 2024;64:2401344. doi: 10.1183/13993003.01344-2024
- Kovacs G, Bartolome S, Denton CP, Gatzoulis MA, Gu S, Khanna D, Badesch D, Montani D. Definition, classification and diagnosis of pulmonary hypertension. *Eur Respir J*. 2024;64:2401324. doi: 10.1183/13993003.01324-2024
- Assad TR, Brittain EL, Wells QS, Farber-Eger EH, Halliday SJ, Doss LN, Xu M, Wang L, Harrell FE, Yu C, et al. Hemodynamic evidence of vascular remodeling in combined post- and precapillary pulmonary hypertension. *Pulm Circ*. 2016;6:313–321. doi: 10.1086/688516
- Assad TR, Hemmes AR, Larkin EK, Glazer AM, Xu M, Wells QS, Farber-Eger EH, Sheng Q, Shyr Y, Harrell FE, et al. Clinical and biological insights into combined post- and pre-capillary pulmonary hypertension. *J Am Coll Cardiol*. 2016;68:2525–2536. doi: 10.1016/j.jacc.2016.09.942
- Huston JH, Shah SJ. Understanding the pathobiology of pulmonary hypertension due to left heart disease. *Circ Res*. 2022;130:1382–1403. doi: 10.1161/CIRCRESAHA.122.319967
- Guazzi M, Ghio S, Adir Y. Pulmonary hypertension in HFpEF and HFrEF: JACC review topic of the week. *J Am Coll Cardiol*. 2020;76:1102–1111. doi: 10.1016/j.jacc.2020.06.069
- Rich S, Rabinovitch M. Diagnosis and treatment of secondary (non-category 1) pulmonary hypertension. *Circulation*. 2008;118:2190–2199. doi: 10.1161/CIRCULATIONAHA.107.723007
- Cody RJ, Haas GJ, Binkley PF, Capers Q, Kelley R. Plasma endothelin correlates with the extent of pulmonary hypertension in patients with chronic congestive heart failure. *Circulation*. 1992;85:504–509. doi: 10.1161/01.cir.85.2.504
- Adir Y, Guazzi M, Offer A, Temporelli PL, Cannito A, Ghio S. Pulmonary hemodynamics in heart failure patients with reduced or preserved ejection fraction and pulmonary hypertension: similarities and disparities. *Am Heart J*. 2017;192:120–127. doi: 10.1016/j.ahj.2017.06.006
- Crawford TC, Leary PJ, Fraser CD, Suarez-Pierre A, Magruder JT, Baumgartner WA, Zehr KJ, Whitman GJ, Masri SC, Sheikh F, et al. Impact of the new pulmonary hypertension definition on heart transplant outcomes: expanding the hemodynamic risk profile. *Chest*. 2020;157:151–161. doi: 10.1016/j.chest.2019.07.028
- Barnett CF, De Marco T. Pulmonary hypertension associated with left-sided heart disease. *Heart Fail Clin*. 2012;8:447–459. doi: 10.1016/j.hfc.2012.04.009
- Fukushi K, Goda A, Kohno T, Takeuchi S, Nakamaru R, Shiraishi Y, Shoji S, Tanaka TD, Nagatomo Y, Kitamura M, et al; West Tokyo Heart Failure Registry Investigators. Revised definition of pulmonary hypertension subtypes enhances prognostic stratification in hospitalized heart failure patients. *J Heart Lung Transplant*. 2026;45:7–15. doi: 10.1016/j.healun.2025.06.009
- Peled Y, Ducharme A, Kittleson M, Bansal N, Stehlik J, Amdani S, Saeed D, Cheng R, Clarke B, Dobbels F, et al. International Society for Heart and Lung Transplantation Guidelines for the evaluation and care of cardiac transplant candidates-2024. *J Heart Lung Transplant*. 2024;43: 1529–1628 e54. doi: 10.1016/j.healun.2024.05.010
- Khush KK, Tasissa G, Butler J, McGlothlin D, De Marco T; ESCAPE Investigators. Effect of pulmonary hypertension on clinical outcomes in advanced heart failure: analysis of the Evaluation Study of Congestive Heart Failure and Pulmonary Artery Catheterization Effectiveness (ESCAPE) database. *Am Heart J*. 2009;157:1026–1034. doi: 10.1016/j.ahj.2009.02.022
- Kilic A, Mathier MA, Hickey GW, Sultan I, Morell VO, Muluksutla SR, Keebler ME. Evolving trends in adult heart transplant with the 2018 heart allocation policy change. *JAMA Cardiol*. 2021;6:159–167. doi: 10.1001/jamacardio.2020.4909
- Masri SC, Tedford RJ, Colvin MM, Leary PJ, Cogswell R. Pulmonary arterial compliance improves rapidly after left ventricular assist device implantation. *ASAIO J*. 2017;63:139–143. doi: 10.1097/MAT.0000000000000467
- Zimpfer D, Zrunek P, Roethy W, Czerny M, Schima H, Huber L, Grimm M, Rajek A, Wolner E, Wieselthaler G. Left ventricular assist devices decrease fixed pulmonary hypertension in cardiac transplant candidates. *J Thorac Cardiovasc Surg*. 2007;133:689–695. doi: 10.1016/j.jtcvs.2006.08.104
- Houston BA, Kalathiya RJ, Hsu S, Loungani R, Davis ME, Coffin ST, Haglund N, Maltais S, Keebler ME, Leary PJ, et al. Right ventricular afterload sensitivity dramatically increases after left ventricular assist device implantation: a multi-center hemodynamic analysis. *J Heart Lung Transplant*. 2016;35:868–876. doi: 10.1016/j.healun.2016.01.1225
- Read JM, Azih NI, Peters CJ, Gurtu V, Vishram-Nielsen JK, Wright SP, Alba AC, Gregoski MJ, Pilch NA, Hsu S, et al. Hemodynamic reserve predicts early right heart failure after LVAD implantation. *J Heart Lung Transplant*. 2022;41:1716–1726. doi: 10.1016/j.healun.2022.07.003
- Cacioli G, Polizzi V, Ciabatti M, Cristiano E, Pergolini A, Distefano G, Lilla Della Monica P, Comisso M, Piazza V, Sbaraglia F, et al. Prediction of right ventricular failure after left ventricular assist device implantation: role of vasodilator challenge. *Eur Heart J Acute Cardiovasc Care*. 2022;11:629–639. doi: 10.1093/ehjacc/zuac085
- Stone JG, Hoar PF, Calabro JR, DePetrillo MA, Bendixen HH. Afterload reduction and preload augmentation improve the anesthetic management of patients with cardiac failure and valvular regurgitation. *Anesth Analg*. 1980;59:737–742
- Lappas DG, Ohtaka M, Fahmy NR, Buckley MJ. Systemic and pulmonary effects of nitroprusside during mitral valve replacement in patients with mitral regurgitation. *Circulation*. 1978;58(3 Pt 2):118–122.
- Pepke-Zaba J, Higenbottam TW, Dinh-Xuan AT, Stone D, Wallwork J. Inhaled nitric oxide as a cause of selective pulmonary vasodilatation in pulmonary hypertension. *Lancet*. 1991;338:1173–1174. doi: 10.1016/0140-6736(91)92033-x
- Wood P, Besterman EM, Towers MK, McIlroy MB. The effect of acetylcholine on pulmonary vascular resistance and left atrial pressure in mitral stenosis. *Br Heart J*. 1957;19:279–286. doi: 10.1136/hrt.19.2.279
- Rich S, Kaufmann E, Levy PS. The effect of high doses of calcium-channel blockers on survival in primary pulmonary hypertension. *N Engl J Med*. 1992;327:76–81. doi: 10.1056/NEJM199207093270203
- Badesch DB, Abman SH, Ahearn GS, Barst RJ, McCrory DC, Simonneau G, McLaughlin VV; American College of Chest Physicians. Medical therapy for pulmonary arterial hypertension: ACCP evidence-based clinical practice guidelines. *Chest*. 2004;126(1 Suppl):35S–62S. doi: 10.1378/chest.126.1_suppl.35S
- Koulova A, Gass AL, Patibandla S, Gupta CA, Aronow WS, Lanier GM. Management of pulmonary hypertension from left heart disease in candidates for orthotopic heart transplantation. *J Thorac Dis*. 2017;9:2640–2649. doi: 10.21037/jtd.2017.07.24
- Rayner SG, Tedford RJ, Leary PJ, Mak S, Houston BA. "This patient needs a doctor, not a guideline": the zone of uncertainty in pulmonary arterial wedge pressure measurement. *Am J Respir Crit Care Med*. 2024;210:712–714. doi: 10.1164/rccm.202402-0359VP
- Humbert M, Kovacs G, Hoeper MM, Badagliacca R, Berger RMF, Brida M, Carlsen J, Coats AJS, Escribano-Subias P, Ferrari P, et al; ESC/ERS Scientific Document Group. 2022 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension. *Eur Heart J*. 2022;43:3618–3731. doi: 10.1093/eurheartj/ehac237
- LeVarge BL, Pomerantsev E, Channick RN. Reliance on end-expiratory wedge pressure leads to misclassification of pulmonary hypertension. *Eur Respir J*. 2014;44:425–434. doi: 10.1183/09031936.00209313
- Viray MC, Bonno EL, Gabrielle ND, Maron BA, Atkins J, Amoroso NS, Fernandes VLC, Maran A, Nielsen CD, Powers ER, et al. Role of pulmonary artery wedge pressure saturation during right heart catheterization: a prospective study. *Circ Heart Fail*. 2020;13:e007981. doi: 10.1161/CIRCHEARTFAILURE.120.007981
- Naeije R, Chin K. Differentiating precapillary from postcapillary pulmonary hypertension. *Circulation*. 2019;140:712–714. doi: 10.1161/CIRCULATIONAHA.119.040295
- Gerhardt F, Fiessler E, Olsson KM, Kayser MZ, Kovacs G, Gall H, Ghofrani HA, Badr Eslam R, Lang IM, Benjamin N, et al. Positive vasoreactivity testing in pulmonary arterial hypertension: therapeutic consequences, treatment patterns, and outcomes in the modern management era. *Circulation*. 2024;149:1549–1564. doi: 10.1161/CIRCULATIONAHA.122.063821
- Belkin MN, Fudim M, Baratto C, Grinstein J, Hollis I, Iijoma N, Kataria R, Lewis GD, Mak S, Tedford RJ, et al; American Heart Association Fellowship-In-Training and Early Career Committee of the Council on Clinical Cardiology. Standardization of baseline and provocative invasive hemodynamic protocols for the evaluation of heart failure and pulmonary hypertension: a scientific statement from the American Heart Association. *Circ Heart Fail*. 2026;19:e000088. doi: 10.1161/HHF.0000000000000088
- Torres Macho J, Delgado Jimenez JF, Sanz Salvo J, Gonzalez Mansilla A, Sanchez Sanchez V, Gamez Diez S, Escibano Subias P, Saez de la Calzada C. Effect of different pharmacologic agents to reverse severe pulmonary hypertension among end-stage heart failure patients. *Transplant Proc*. 2009;41:2477–2479. doi: 10.1016/j.transproceed.2009.06.054

37. Givertz Michael M, et al. Effect of bolus Milrinone on hemodynamic variables and pulmonary vascular resistance in patients with severe left ventricular dysfunction: a rapid test for reversibility of pulmonary hypertension. *JACC*. 1996;28:1775–1780. doi: 10.1016/S0735-1097(96)00399-3
38. Sablotzki A, Hentschel T, Gruenig E, Schubert S, Friedrich I, Mühling J, Dehne MG, Czeslick E. Hemodynamic effects of inhaled aerosolized iloprost and inhaled nitric oxide in heart transplant candidates with elevated pulmonary vascular resistance. *Eur J Cardio-thorac Surg*. 2002;22:746–752. doi: 10.1016/s1010-7940(02)00488-8
39. Sablotzki A, Hentschel T, Hofmann S, Simm A, Mühling J, Czeslick E. Inhaled aerosolized iloprost in the evaluation of heart transplant candidates—experiences with 45 cases. *J Clin Anesth*. 2006;18:108–113. doi: 10.1016/j.jclinane.2005.08.018
40. Strong C, Raposo L, Castro M, Madeira S, Tralhão A, Ventosa A, Rebocho MJ, Almeida M, Aguiar C, Neves JP, et al. Haemodynamic effects and potential clinical implications of inhaled nitric oxide during right heart catheterization in heart transplant candidates. *ESC Heart Fail*. 2020;7:673–681. doi: 10.1002/ehf2.12639
41. Guglin M, Mehra S, Mason TJ. Comparison of drugs for pulmonary hypertension reversibility testing: A meta-analysis. *Pulm Circ*. 2013;3:406–413. doi: 10.4103/2045-8932.113180
42. Guazzi M. Pulmonary hypertension and heart failure: a dangerous liaison. *Heart Fail Clin*. 2018;14:297–309. doi: 10.1016/j.hfc.2018.02.006
43. Preston IR, Sagliani KD, Roberts KE, Shah AM, Desouza SA, Howard W, Brennan J, Hill NS. Comparison of acute hemodynamic effects of inhaled nitric oxide and inhaled epoprostenol in patients with pulmonary hypertension. *Pulm Circ*. 2013;3:68–73. doi: 10.4103/2045-8932.109916
44. Goland S, Czer LSC, Kass RM, De Robertis MA, Mirocha J, Coleman B, Capelli C, Raissi S, Cheng W, Fontana G, et al. Pre-existing pulmonary hypertension in patients with end-stage heart failure: impact on clinical outcome and hemodynamic follow-up after orthotopic heart transplantation. *J Heart Lung Transplant*. 2007;26:312–318. doi: 10.1016/j.healun.2006.12.012
45. Drakos SG, Kfoury AG, Gilbert EM, Horne BD, Long JW, Stringham JC, Campbell BA, Renlund DG. Effect of reversible pulmonary hypertension on outcomes after heart transplantation. *J Heart Lung Transplant*. 2007;26:319–323. doi: 10.1016/j.healun.2007.01.012
46. Ghio S, Crimi G, Pica S, Temporelli PL, Boffini M, Rinaldi M, Raineri C, Scelsi L, Pistono M, Totaro R, et al. Persistent abnormalities in pulmonary arterial compliance after heart transplantation in patients with combined post-capillary and pre-capillary pulmonary hypertension. *PLoS One*. 2017;12:e0188383. doi: 10.1371/journal.pone.0188383
47. Bhatia SJ, Kirshenbaum JM, Shemin RJ, Cohn LH, Collins JJ, Di Sesa VJ, Young PJ, Mudge GH, Sutton MG. Time course of resolution of pulmonary hypertension and right ventricular remodeling after orthotopic cardiac transplantation. *Circulation*. 1987;76:819–826. doi: 10.1161/01.cir.76.4.819
48. Colvin MM, Smith JM, Stomberski CT, Ahn YS, Lindblad KA, Messick EH, Handarova D, Hart A, Snyder JJ. OPTN/SRTR 2024 annual data report: heart. *Am J Transplant*. 2026;26:S367–S473. doi: 10.1016/j.ajt.2026.05.723
49. Costard-Jackle A, Fowler MB. Influence of preoperative pulmonary artery pressure on mortality after heart transplantation: testing of potential reversibility of pulmonary hypertension with nitroprusside is useful in defining a high risk group. *J Am Coll Cardiol*. 1992;19:48–54. doi: 10.1016/0735-1097(92)90050-w
50. Murali S, Kormos RL, Uretsky BF, Schechter D, Reddy PS, Denys BG, Armitage JM, Hardesty RL, Griffith BP. Preoperative pulmonary hemodynamics and early mortality after orthotopic cardiac transplantation: the Pittsburgh experience. *Am Heart J*. 1993;126:896–904. doi: 10.1016/0002-8703(93)90704-d
51. Delgado JF, Gómez-Sánchez MA, Sáenz de la Calzada C, Sánchez V, Escribano P, Hernández-Afonso J, Tello R, Gómez de la Cámara A, Rodríguez E, Rufilanchas JJ. Impact of mild pulmonary hypertension on mortality and pulmonary artery pressure profile after heart transplantation. *J Heart Lung Transplant*. 2001;20:942–948. doi: 10.1016/s1053-2498(01)00286-8
52. Klotz S, Deng MC, Hanafy D, Schmid C, Stypmann J, Schmidt C, Hammel D, Scheld HH. Reversible pulmonary hypertension in heart transplant candidates—pretransplant evaluation and outcome after orthotopic heart transplantation. *Eur J Heart Fail*. 2003;5:645–653. doi: 10.1016/s1388-9842(03)00059-x
53. Klotz S, Wenzelburger F, Stypmann J, Welp H, Drees G, Schmid C, Scheld HH. Reversible pulmonary hypertension in heart transplant candidates: to transplant or not to transplant. *Ann Thorac Surg*. 2006;82:1770–1773. doi: 10.1016/j.athoracsurg.2006.05.114
54. Butler J, Stankewicz MA, Wu J, Chomsky DB, Howser RL, Khadim G, Davis SF, Pierson RN, Wilson JR. Pre-transplant reversible pulmonary hypertension predicts higher risk for mortality after cardiac transplantation. *J Heart Lung Transplant*. 2005;24:170–177. doi: 10.1016/j.healun.2003.09.045
55. Vakil K, Duval S, Sharma A, Adabag S, Abidi KS, Taimhe Z, Colvin-Adams M. Impact of pre-transplant pulmonary hypertension on survival after heart transplantation: a UNOS registry analysis. *Int J Cardiol*. 2014;176:595–599. doi: 10.1016/j.ijcard.2014.08.072
56. Rzechorzek W, Dey S, Shah A, Michaud L, Boyle D, Pitaktong A, Levine A, Pan S, Ohira S, Gass A, et al. High-risk pulmonary hypertension does not worsen outcomes in heart transplantation. *Am J Cardiol*. 2023;205:290–297. doi: 10.1016/j.amjcard.2023.07.009
57. Rivinius R, Helmschrott M, Ruhparwar A, Schmack B, Darche FF, Thomas D, Bruckner T, Doesch AO, Katus HA, Ehlermann P. Elevated pre-transplant pulmonary vascular resistance is associated with early post-transplant atrial fibrillation and mortality. *ESC Heart Fail*. 2020;7:176–187. doi: 10.1002/ehf2.12549
58. Lim HS, Zaphiriou A. Sodium nitroprusside in patients with mixed pulmonary hypertension and left heart disease: hemodynamic predictors of response and prognostic implications. *J Card Fail*. 2016;22:117–124. doi: 10.1016/j.cardfail.2015.10.018
59. Satoh T, Yaoita N, Nochioka K, Tatebe S, Hayashi H, Yamamoto S, Sato H, Takahama H, Suzuki H, Terui Y, et al. Inhaled nitric oxide testing in predicting prognosis in pulmonary hypertension due to left-sided heart diseases. *ESC Heart Fail*. 2023;10:3592–3603. doi: 10.1002/ehf2.14515
60. Ghio S, Crimi G, Houston B, Motalto C, Garascia A, Boffini M, Temporelli PL, La Rovere MT, Pacileo G, Panneerselvam K, et al. Nonresponse to acute vasodilator challenge and prognosis in heart failure with pulmonary hypertension. *J Card Fail*. 2021;27:869–876. doi: 10.1016/j.cardfail.2021.01.021
61. von Scheidt W, Costard-Jackle A, Stempfle HU, Deng MC, Schwaab B, Haaff B, Naegele H, Mohacs P, Trautnitz M, PROPHET Study Group. Prostaglandin E1 testing in heart failure associated pulmonary hypertension enables transplantation: the PROPHET study. *J Heart Lung Transplant*. 2006;25:1070–1076. doi: 10.1016/j.healun.2006.04.011
62. Ilonze OJ, Ebong IA, Guglin M, Nair A, Rich J, McLaughlin V, Tedford RJ, Mazimba S. Considerations in the diagnosis and management of pulmonary hypertension associated with left heart disease. *JACC Heart Fail*. 2024;12:1328–1342. doi: 10.1016/j.jchf.2024.04.031
63. Sharma S, Ruiz J, Paghdar S, Desai S, Goswami R. Revisiting pulmonary hypertension in the era of temporary mechanical circulatory support - literature review and case-based discussion. *Transplant Proc*. 2023;55:2462–2469. doi: 10.1016/j.transproceed.2023.09.022
64. Haddad O, Sareyyupoglu B, Goswami RM, Bitargil M, Patel PC, Jacob S, El-Sayed Ahmed MM, Leoni Moreno JC, Yip DS, Landolfo K, et al. Short-term outcomes of heart transplant patients bridged with Impella 5.5 ventricular assist device. *ESC Heart Fail*. 2023;10:2298–2306. doi: 10.1002/ehf2.14391
65. Everett KD, Jain P, Botto R, Salama M, Miyashita S, Reyelt L, Kawabori M, Kapur NK. Acute effects of left ventricular support with Impella 5.5 on biventricular hemodynamics. *Circ Heart Fail*. 2021;14:e008616. doi: 10.1161/CIRCHEARTFAILURE.121.008616
66. Kabirpour A, Saad M, Aziz M, Itare V, Ochoa RJ, Sharma A, Callicurn K, Wheeler J, Itagaki S, Cagliostro M, et al. Extent of pulmonary vascular resistance reduction with temporary mechanical circulatory devices among patients awaiting cardiac transplantation. *J Heart Lung Transplant*. 2024;43:S238–S239. doi: 10.1016/j.healun.2024.02.1124
67. Rankin J, Imielski B, Howard-Green A, Saxena A, Raman D, Seals A, Paik K, Patel P, Mishkin J, Adeniyi A, et al. Improvement in pulmonary vascular resistance with Impella 5.5 as a bridge to orthotopic heart transplantation. *J Heart Lung Transplant*. 2026;45:46–47. doi: 10.1016/j.healun.2026.02.104
68. Garg P, Alomari M, Wadiwala I, Lyle M, Pham S, Celik NB, Moreno JCL, Goswami R, Landolfo K, Nativi-Nicolau J, et al. High profile transvalvular microaxial flow pump as a bridge to heart transplantation for amyloid cardiomyopathy: a case series. *JHLT Open*. 2025;10:100340. doi: 10.1016/j.jhlto.2025.100340
69. Ruiz J, Desai S, Paghdar S, Malkani S, Nativi J, Yip D, Patel P, Leoni J, Lyle M, Goswami R. (807) the impact of axillary mechanical circulatory support in patients awaiting heart transplantation with pulmonary hypertension. *J Heart Lung Transplant*. 2023;42:S354. doi: 10.1016/j.healun.2023.02.820
70. Acuna-Morin EE, Blumer V, Rivera-Maza M, Kittipibul V, Maning J, Chaparro S, Joyce E. Reversibility of fixed pulmonary hypertension with LVADS as a bridge to candidacy strategy for heart transplantation: a systematic review and meta-analysis. *J Card Fail*. 2019;25(8, Supplement):S160. doi: 10.1016/j.cardfail.2019.07.459
71. Anegawa E, Seguchi O, Mochizuki H, Kuroda K, Doi Nakajima S, Watanabe T, Yanase M, Tadokoro N, Fukushima S, Fujita T, et al. Pulmonary vascular reverse remodeling after left ventricular assist device implantation

- in patients with pulmonary hypertension. *ASAIO J*. 2023;69:151–158. doi: 10.1097/MAT.0000000000001739
72. Alba AC, Rao V, Ross HJ, Jensen AS, Sander K, Gustafsson F, Delgado DH. Impact of fixed pulmonary hypertension on post-heart transplant outcomes in bridge-to-transplant patients. *J Heart Lung Transplant*. 2010;29:1253–1258. doi: 10.1016/j.healun.2010.06.002
 73. Grupper A, Mazin I, Faierstein K, Kurnick A, Maor E, Elian D, Barbash IM, Guetta V, Regev E, Morgan A, et al. Hemodynamic changes after left ventricular assist device implantation among heart failure patients with and without elevated pulmonary vascular resistance. *Front Cardiovasc Med*. 2022;9:875204. doi: 10.3389/fcvm.2022.875204
 74. John R, Liao K, Kamdar F, Eckman P, Boyle A, Colvin-Adams M. Effects on pre- and posttransplant pulmonary hemodynamics in patients with continuous-flow left ventricular assist devices. *J Thorac Cardiovasc Surg*. 2010;140:447–452. doi: 10.1016/j.jtcvs.2010.03.006
 75. Kutty RS, Parameshwar J, Lewis C, Catarino PA, Sudarshan CD, Jenkins DP, Dunning JJ, Tsui SS. Use of centrifugal left ventricular assist device as a bridge to candidacy in severe heart failure with secondary pulmonary hypertension. *Eur J Cardio-Thorac Surg*. 2013;43:1237–1242. doi: 10.1093/ejcts/ezs678
 76. Mikus E, Stepanenko A, Krabatsch T, Loforte A, Dandel M, Lehmkuhl HB, Hetzer R, Potapov EV. Reversibility of fixed pulmonary hypertension in left ventricular assist device support recipients. *Eur J Cardiothorac Surg*. 2011;40:971–977. doi: 10.1016/j.ejcts.2011.01.019
 77. Selim AM, Wadhvani L, Burdorf A, Raichlin E, Lowes B, Zolty R. Left ventricular assist devices in pulmonary hypertension group 2 with significantly elevated pulmonary vascular resistance: a bridge to cure. *Heart Lung Circ*. 2019;28:946–952. doi: 10.1016/j.hlc.2018.04.299
 78. Hoepfer MM, Oerke B, Wissmüller M, Leuchte H, Opitz C, Halank M, Seyfarth H-J, Baldus S, Bauersachs J, Böhm M, et al. Tadalafil for treatment of combined postcapillary and precapillary pulmonary hypertension in patients with heart failure and preserved ejection fraction: a randomized controlled phase 3 study. *Circulation*. 2024;150:600–610. doi: 10.1161/CIRCULATIONAHA.124.069340
 79. Cooper TJ, Cleland JGF, Guazzi M, Pellicori P, Ben Gal T, Amir O, Al-Mohammad A, Clark AL, McConnell A, Steine K, et al. Effects of sildenafil on symptoms and exercise capacity for heart failure with reduced ejection fraction and pulmonary hypertension (the Sildenafil study): a randomized placebo-controlled multicentre trial. *Eur J Heart Fail*. 2022;24:1239–1248. doi: 10.1002/ehfj.2527
 80. Bermejo J, Yotti R, García-Orta R, Sánchez-Fernández PL, Castaño M, Segovia-Cubero J, Escribano-Subías P, San Román JA, Borrás X, Alonso-Gómez A, et al; Sildenafil for Improving Outcomes after Valvular Correction (SIOVAC) investigators. Sildenafil for improving outcomes in patients with corrected valvular heart disease and persistent pulmonary hypertension: a multicenter, double-blind, randomized clinical trial. *Eur Heart J*. 2018;39:1255–1264. doi: 10.1093/eurheartj/ehx700
 81. Shah SJ, Bonderman D, Borlaug BA, Cleland JGF, Lack G, Lu W, Voors AA, Zannad F, Gladwin MT. Macitentan for heart failure with preserved or mildly reduced ejection fraction and pulmonary vascular disease: results of the SERENADE randomized clinical trial and open-label extension study. *Circ Heart Fail*. 2025;18:e011381. doi: 10.1161/CIRCHEARTFAILURE.123.011381
 82. Vachieri JL, Delcroix M, Al-Hiti H, Efficace M, Hutrya M, Lack G, Papadakis K, Rubin LJ, Macitentan in pulmonary hypertension due to left ventricular dysfunction. *Eur Respir J*. 2018;51:1701886. doi: 10.1183/13993003.01886-2017
 83. Packer M, McMurray JJV, Krum H, Kiowski W, Massie BM, Caspi A, Pratt CM, Petrie MC, DeMets D, Kober L, et al; ENABLE Investigators and Committees. Long-term effect of endothelin receptor antagonism with Bosentan on the morbidity and mortality of patients with severe chronic heart failure: primary results of the ENABLE trials. *JACC Heart Fail*. 2017;5:317–326. doi: 10.1016/j.jchf.2017.02.021
 84. Dachs TM, Duca F, Retzl R, Binder-Rodriguez C, Dalos D, Ligios LC, Kammerlander A, Grünig E, Pretsch I, Steringer-Mascherbauer R, et al. Riociguat in pulmonary hypertension and heart failure with preserved ejection fraction: the haemoDYNAMIC trial. *Eur Heart J*. 2022;43:3402–3413. doi: 10.1093/eurheartj/ehac389
 85. García-Alvarez A, Blanco I, García-Lunar I, Jordà P, Rodríguez-Arias JJ, Fernández-Friera L, Zegri I, Nuche J, Gomez-Bueno M, Prat S, et al; SPHERE-HF Investigators. beta3 adrenergic agonist treatment in chronic pulmonary hypertension associated with heart failure (SPHERE-HF): a double blind, placebo-controlled, randomized clinical trial. *Eur J Heart Fail*. 2023;25:373–385. doi: 10.1002/ehfj.2745
 86. Fayyaz AU, Edwards WD, Maleszewski JJ, Konik EA, DuBrock HM, Borlaug BA, Frantz RP, Jenkins SM, Redfield MM. Global pulmonary vascular remodeling in pulmonary hypertension associated with heart failure and preserved or reduced ejection fraction. *Circulation*. 2018;137:1796–1810. doi: 10.1161/CIRCULATIONAHA.117.031608
 87. Pons J, Leblanc M-H, Bernier M, Cantin B, Bourgault C, Bergeron S, Proulx G, Morin J, Nalli C, O'Connor K, et al. Effects of chronic sildenafil use on pulmonary hemodynamics and clinical outcomes in heart transplantation. *J Heart Lung Transplant*. 2012;31:1281–1287. doi: 10.1016/j.healun.2012.09.009
 88. De Santo LS, Romano G, Maiello C, Buonocore M, Cefarelli M, Galdieri N, Nappi G, Amarelli C. Pulmonary artery hypertension in heart transplant recipients: How much is too much? *Eur J Cardiothorac Surg*. 2012;42:864–9; discussion 869. doi: 10.1093/ejcts/ezs102
 89. Reichenbach A, Al-Hiti H, Malek I, Pirk J, Goncalvesova E, Kautzner J, Melenovsky V. The effects of phosphodiesterase 5 inhibition on hemodynamics, functional status and survival in advanced heart failure and pulmonary hypertension: a case-control study. *Int J Cardiol*. 2013;168:60–65. doi: 10.1016/j.ijcard.2012.09.074
 90. Gombert-Maitland M, Tedford RJ, Langleben D, Rosenkranz S, Miller B, Jones AD, Urbinati A, McMullan CJ, Cornell AG, Vachieri J-L. Sotatercept for combined post- and precapillary pulmonary hypertension associated with heart failure: results from the phase 2, randomized, placebo-controlled CADENCE study. *Circulation*. 2026;153:1446–1459. doi: 10.1161/CIRCULATIONAHA.126.079918
 91. Nassif ME, Qintar M, Windsor SL, Jermyn R, Shavelle DM, Tang F, Lamba S, Bhatt K, Brush J, Civitello A, et al. Empagliflozin effects on pulmonary artery pressure in patients with heart failure: results from the EMBRACE-HF trial. *Circulation*. 2021;143:1673–1686. doi: 10.1161/CIRCULATIONAHA.120.052503
 92. Borlaug BA, Reddy YNV, Braun A, Sorin M, Sorinachi H, Omar M, Popovic D, Alogna A, Jensen MD, Carter R. Cardiac and metabolic effects of dapagliflozin in heart failure with preserved ejection fraction: the CAMEO-DAPA trial. *Circulation*. 2023;148:834–844. doi: 10.1161/CIRCULATIONAHA.123.065134
 93. Han Y, Cho Y-E, Ayon R, Guo R, Yousef KD, Pan M, Dai A, Yuan JX-J, Makino A. SGLT inhibitors attenuate NO-dependent vascular relaxation in the pulmonary artery but not in the coronary artery. *Am J Physiol Lung Cell Mol Physiol*. 2015;309:L1027–L1036. doi: 10.1152/ajplung.00167.2015
 94. Chowdhury B, Luu AZ, Luu VZ, Kabir MG, Pan Y, Teoh H, Quan A, Sabongui S, Al-Omran M, Bhatt DL, et al. The SGLT2 inhibitor empagliflozin reduces mortality and prevents progression in experimental pulmonary hypertension. *Biochem Biophys Res Commun*. 2020;524:50–56. doi: 10.1016/j.bbrc.2020.01.015
 95. Masarone D, Valente F, Verriglia M, Contaldi C, di Palma V, Falco L, Catapano D, di Lorenzo E. Pharmacologic treatment of pulmonary hypertension due to heart failure with preserved ejection fraction: Are there more arrows on our bow? *J Clin Med*. 2024;13:6867. doi: 10.3390/jcm13226867
 96. Zhang J, Du Le, Qin X, Guo X. Effect of sacubitril/valsartan on the right ventricular function and pulmonary hypertension in patients with heart failure with reduced ejection fraction: a systematic review and meta-analysis of observational studies. *J Am Heart Assoc*. 2022;11:e024449. doi: 10.1161/JAHA.121.024449
 97. Burdorf C, Brockmüller J, Strampe H, Januszewski M, Remppis BA. Reduction of pulmonary hypertension after transition to sacubitril/valsartan in patients with heart failure with preserved ejection fraction. *Front Cardiovasc Med*. 2021;8:734697. doi: 10.3389/fcvm.2021.734697
 98. Codina P, Domingo M, Barceló E, Gastelurrutia P, Casquete D, Vila J, Abdul-Jawad Altisent O, Spitaleri G, Cediel G, Santiago-Vacas E, et al. Sacubitril/valsartan affects pulmonary arterial pressure in heart failure with preserved ejection fraction and pulmonary hypertension. *ESC Heart Fail*. 2022;9:2170–2180. doi: 10.1002/ehf2.13952
 99. Shah M, Saeed O, Shin J, Murthy S, Sims DB, Vukelic S, Goldstein D, Forest SJ, Jorde UP, Patel SR. Predicted heart mass-based size matching among recipients with moderate pulmonary hypertension: Outcomes and sex effect. *J Heart Lung Transplant*. 2020;39:648–656. doi: 10.1016/j.healun.2020.01.1339
 100. Jedeon Z, Pillai A, Baker WL, Jaiswal A. Donor pulmonary hemodynamics does not impact recipient outcomes in adult heart transplantation. *Int J Cardiol*. 2025;420:132747. doi: 10.1016/j.ijcard.2024.132747